

Performance Testing Report

Date	07 July 2026
Project Name	CaptionAI: AI Powered Social Media Caption Generator
Maximum Marks	5 Marks

Performance Testing Report:

Performance testing was carried out to ensure CaptionAI is responsive, stable, and efficient under normal operating conditions. As an individually developed academic project, testing was performed manually through the browser rather than with dedicated load-testing tools; the results below reflect single-user, real-world usage observations rather than automated multi-user simulations.

Step 1: Test Scenarios & Tooling

S.No	Test Type	Scenario / API Endpoint Tested	Tool(s) Used
1	Load Testing	Multiple sequential AI caption generation requests were made to simulate typical repeated usage by a single user.	Manual testing via Google Chrome browser (no dedicated load-testing tool used)
2	Stress Testing	Repeated caption generation requests were sent in succession to observe application stability and the Groq API's behavior under continuous use.	Manual testing via Google Chrome browser (no dedicated stress-testing tool used)
3	Spike Testing	Not performed – formal spike/sudden-burst traffic testing was outside the scope of this individually developed academic project.	Not applicable
4	Endurance Testing	Not performed – no extended (e.g., 24-hour) continuous run was conducted to check for memory leaks.	Not applicable

Step 2: Key Performance Indicators (KPIs) & Results

No formal SLA-style thresholds were pre-defined for this academic project; the targets below reflect general, acceptable web-application UX benchmarks used to evaluate the measured results.

S.No	Metric (KPI)	Target Threshold	Actual Result	Status
1	Average Response Time	< 2 seconds	Login response: < 2 seconds; Page loading: approx. 1–2 seconds	Pass
2	Peak Response Time (Caption Generation)	< 6 seconds	Approx. 3–5 seconds, depending on Groq API response time	Pass
3	Throughput (Requests/sec)	Not formally targeted	Not measured – testing was single-user and manual; no concurrent load was simulated	Not Measured

S.No	Metric (KPI)	Target Threshold	Actual Result	Status
4	Error Rate	< 1%	0% – no crashes or failures observed across all test cases	Pass
5	CPU / Memory Utilization	< 80% under load	CPU: approx. 15–30% during caption generation; Memory: approx. 180–250 MB	Pass

Test Environment

Operating System	Windows 11
Processor	Intel Core i5
RAM	8 GB
Development Environment	Visual Studio Code
Backend	Python Flask
Database	SQLite
AI Model	Groq API (LLaMA 3.3-70B Versatile)
Browser	Google Chrome
Internet Connection	Stable Broadband (50 Mbps)

Functional Test Cases

Feature Tested	Expected Result	Actual Result	Status
User Registration	User account should be created successfully.	Account created successfully.	Passed
User Login	User should log in using valid credentials.	Login successful.	Passed
AI Caption Generation	Generate captions, hashtags, and CTA.	AI generated accurate content within 3–5 seconds.	Passed
Caption History	Generated captions should be stored and displayed.	History displayed correctly.	Passed
Copy & Download	Users should be able to copy or download captions.	Copy and download functions worked correctly.	Passed

Stress Testing Observations

The application was tested with multiple caption generation requests in succession.

- Application remained stable.
- No crashes were observed.
- Database operations completed successfully.
- AI responses were received consistently.
- Slight increase in response time during repeated API requests due to internet latency.

Issues Identified

- AI response time depends on internet connectivity.
- API requests may take longer during high network latency.
- SQLite is suitable for this academic project but may not be ideal for very large-scale deployments.

Improvements Implemented

- Optimized Flask routes.
- Improved database queries.
- Enhanced input validation.
- Reduced unnecessary page reloads.
- Improved error handling for API failures.

Conclusion

The CaptionAI application performed reliably during testing. All major functionalities, including user authentication, AI caption generation, hashtag generation, CTA suggestions, caption history, and download features, worked as expected. Response times were satisfactory for an academic web application, and the system remained stable throughout the testing process.